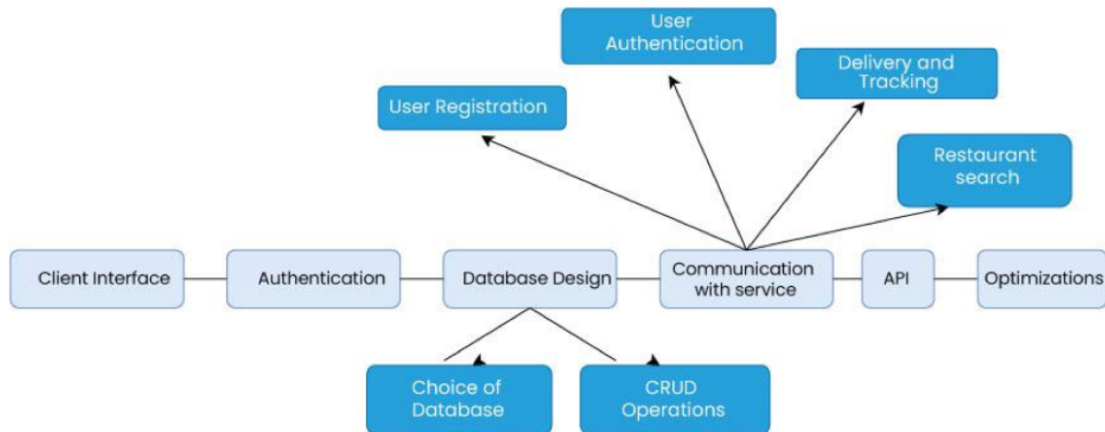


Low Level Design

LLD (Low Level Design)



Database Design for Restaurant Management System

1. Customer

```
{
  -customer_Id
  -customer_name,
  -address
  , -phone
  _no,
  -email
}
```

- **Customer_Id (PK):** Unique identifier for each customer.
- **Customer_name:** Stores the name of the customer.
- **Address:** Records the address of the customer.
- **Phone_no:** Stores the phone number of the customer.
- **Email:** Captures the email address of the customer.

2. Restaurant

```
{
  -restaurant_id,
  -restaurant_name,
  -payment_details_id
}
```

```
-payment_details_id,
-location,
-menu_Id
}
```

- **Restaurant_id (PK):** Unique identifier for each restaurant.
- **Restaurant_name:** Stores the name of the restaurant.
- **Payment_details_id (FK):** Foreign key linked to the payment details table for crediting transactions.
- **Location:** Records the location of the restaurant.
- **Menu_Id (FK):** Foreign key pointing to the menu table for menu modifications by the restaurant admin.

3. Delivery agent

```
{
  -restaurant_id,
  -restaurant_name,
  -payment_details_id,
  -location,
  -menu_Id
}
```

- **Agent_id (PK):** Unique identifier for each delivery agent.
- **Payment_details_id (FK):** Foreign key linked to the payment details table for agent's account information.
- **Phone_no:** Stores the phone number of the delivery agent.
- **Email:** Captures the email address of the delivery agent.
- **Vehicle_number:** Records the vehicle number used by the agent for deliveries.
- **Address:** Stores the permanent address of the delivery agent.

4. Order

```
{
  -order_id,
  -customer_id,
  -restaurant_id,
  -agent_id,
  -time_stamp,
  -is_delivery,
  -delivery_location,
  -items_id,
  -delivery_price,
  -items_price,
  -transcation_id:
  -order_status
}
```

- **Order_id (PK):** Unique identifier for each order.
- **Customer_id (FK):** Foreign key linking to the customer table.
- **Restaurant_id (FK):** Foreign key linking to the restaurant table.
- **Agent_id (FK):** Foreign key linking to the delivery agent table (NULL for dine-in orders)

- **Agent_id (FK):** Foreign key linking to the delivery agent table (NULL for dine-in orders).
- **Time_stamp:** Records the timestamp when the order was placed.
- **Is_delivery:** Boolean value indicating whether the order is for delivery (true) or dine-in (false).
- **Delivery_location:** Stores the location for delivery orders (NULL for dine-in).
- **Items_id (FK):** Foreign key pointing to the table storing items and quantities.
- **Delivery_price:** Records the delivery charges calculated by the Delivery Services based on delivery proximity.
- **Items_price:** Stores the total sum of the price of ordered items.
- **Order_status:** Records the status of the order (e.g., delivered, cancelled).
- **Transaction_id:** Stores the transaction details for the specific order.

NoSQL data

1. Restaurant_rating

```
{
  -restaurant_id,
  -rating
}
```

- **Restaurant_id:** Unique identifier for each restaurant.
- **Rating:** Stores the rating given to the restaurant.

2. Agent_rating

```
{
  -agent_id,
  -rating,
}
```

- **Agent_id:** Unique identifier for each delivery agent.
- **Rating:** Records the rating given to the delivery agent.

Tomorrow I will explore further about Database Design